

# Phase Admissibility and the Global Consistency Functional

**Marlon Hanks**

Independent Researcher

Dust LLC

*Subject Classification: Foundations of Physics · Unified Field Theory · Quantum Gravity*

**Keywords:** *phase admissibility, consistency functional, global coherence, topological stability, emergent dynamics, action principle, phase ontology*

## Abstract

We introduce the phase admissibility functional  $\mathcal{I}[\Phi]$ , a global consistency measure that determines which phase configurations are physically realizable within Phase Theory. This functional replaces the roles traditionally played by action principles, equations of motion, and probabilistic postulates. Physical evolution is defined as admissible transformation of phase configurations that preserve or extremize  $\mathcal{I}$ . We establish the formal axiomatic structure of  $\mathcal{I}[\Phi]$ , prove that discreteness and stability arise necessarily from admissibility constraints alone, derive the emergence of an effective action principle in appropriate limits, and show that topological protection of stable structures follows from the coherence and topological terms in the functional's canonical decomposition. Forbidden configurations are shown to be ontologically nonexistent rather than dynamically suppressed. We enumerate formal requirements on  $\mathcal{I}$ , prove key structural theorems, compare Phase Theory's framework to classical and quantum action principles, and specify falsifiability conditions. This paper establishes the operational and mathematical core of Phase Theory, bridging the foundational ontology of Paper 1

with the spectral, gravitational, and measurement-theoretic developments of subsequent papers.

---

## 1. Introduction and Motivation

The two pillars of twentieth-century theoretical physics—quantum mechanics and general relativity—have resisted unification for nearly a century. Despite the extraordinary predictive successes of each framework within its domain, their fundamental incompatibility at the Planck scale suggests that both are effective approximations of a deeper, as-yet-unidentified theoretical structure. String theory, loop quantum gravity, causal set theory, and related programs have each sought this structure by modifying or extending one or both pillars, yet no program has achieved a derivation of the full phenomenology of the Standard Model and gravitation from a single, parsimonious primitive. Paper 1 of this series [1] established a radically different starting point: that physical reality reduces to a single primitive entity, *phase*, denoted  $(\Phi)$ , defined as a map from a base topological space into a target phase manifold. Spacetime, matter, fields, geometry, and all emergent structure are encoded in the global organization of  $(\Phi)$ . The present paper, Paper 2 of the Phase Theory series, takes the foundational ontology of Paper 1 as its point of departure and constructs the operational mechanism that animates that ontology: the global phase admissibility functional  $(\mathcal{I}[\Phi])$ .

The mere existence of a primitive, however well-motivated ontologically, is insufficient to constitute a complete physical theory. A theory must explain at minimum three things: (a) why some phase configurations are realized and others are not, given that the configuration space  $(\mathcal{C})$  is vast; (b) why realized structures are stable over extended parameter ranges rather than fleeting or ephemeral; and (c) why physical evolution appears ordered and lawful rather than arbitrary, chaotic, or reversible without distinction.

Traditional physics addresses these three requirements separately, and by means of exogenous postulates: equations of motion select trajectories from the space of all field configurations; an action functional, extremized according to Hamilton's principle, generates these equations; and probabilistic postulates—the Born rule in quantum mechanics, statistical mechanics in thermodynamics—assign physical significance to the spectrum of possibilities. Phase Theory proposes a unified replacement for all three mechanisms: the global admissibility functional  $\mathcal{I}[\Phi]$ , a single object whose structure simultaneously accounts for realizability, stability, and lawfulness. The conceptual economy of this replacement is not merely aesthetic—it is demanded by the ontological parsimony of Phase Theory, which prohibits exogenous postulates as violations of the principle that  $\Phi$  is the sole primitive.

Without a selection criterion, the phase substrate would support all configurations in  $\mathfrak{C}$  equally, yielding a maximally disordered universe inconsistent with observation. The persistence of macroscopic structure—stable particles, long-range fields, geometric order, thermodynamic asymmetry—implies that extremely strong constraints operate on the set of configurations that are physically realized. These constraints must be intrinsic to  $\Phi$  itself, not externally imposed. To impose them externally would be to introduce a second primitive—the constraint mechanism—in addition to phase, violating the foundational commitment of the theory. The admissibility functional is the minimal additional object that can implement intrinsic selection: it is a functional on the configuration space, defined by properties of  $\Phi$  alone, that partitions  $\mathfrak{C}$  into the realizable and the nonexistent. It is important to note at the outset what  $\mathcal{I}[\Phi]$  is not. It is not an action in the Lagrangian sense—it does not require a background spacetime metric over which to integrate, it is not extremized along a parameterized trajectory, and it carries no probabilistic interpretation whatsoever. It is a global object: a

map from the space of all phase configurations to the real line, whose sign determines the ontological status of each configuration. The equations of motion, the action principle, and the Born rule each emerge as effective descriptions valid in particular limiting regimes of the admissibility criterion, as subsequent sections demonstrate.

The present paper serves a precise structural role within the Phase Theory series. Papers 3 through 9 derive, respectively: the discrete spectrum of stable particles, the propagation and polarization structure of phase photons, emergent quantum statistics and the measurement problem, gauge structure and confinement, gravitational dynamics from the emergent action, and cosmological implications including the origin of temporal order. All of these derivations depend fundamentally on the existence, axioms, and canonical decomposition of  $\mathcal{I}[\Phi]$  established in the present work. Paper 2 is therefore the operational core of Phase Theory—the bridge between the foundational ontology of Paper 1 and the rich physical derivations of Papers 3–9. The reader who accepts the axioms and formal results of this paper will find that the subsequent papers follow from them by mathematical necessity rather than additional postulation. The principal claim of the present paper may be stated succinctly: given phase as the sole ontological primitive and  $\mathcal{I}[\Phi]$  as defined herein, the full structure of physical law—dynamics, stability, discreteness, causal order, and probabilistic appearance—is derivable without further assumption.

The paper is organized as follows. Section 2 reviews the foundational setup of Paper 1. Section 3 argues for the necessity of an admissibility functional. Section 4 provides the formal definition of  $\mathcal{I}[\Phi]$  via axioms A1–A5. Section 5 presents the canonical three-term decomposition of  $\mathcal{I}$ . Sections 6 and 7 analyze the coherence and topological terms in detail, proving key lemmas and theorems. Section 8 formalizes admissible evolution. Section 9 derives the emergence of the action principle. Section 10

establishes ontological elimination. Section 11 derives spectral discreteness. Section 12 treats the emergence of probability. Section 13 compares Phase Theory with classical and quantum action principles. Section 14 compares with competing quantum gravity programs. Section 15 specifies the role of this paper's results for Papers 3–9. Section 16 states falsifiability conditions. Section 17 analyzes the admissibility boundary and phase transitions. Section 18 characterizes the mathematical structure of the admissible configuration space. Section 19 demonstrates consistency with established physics. Section 20 concludes.

## 2. Review of Paper 1: Phase Primacy

Paper 1 of this series established the foundational ontology of Phase Theory [1]. The central claim is that physical reality admits a complete description in terms of a single entity: *phase*. Formally, phase is defined as a map

$$(\Phi: \mathcal{B} \rightarrow \mathcal{M}_\phi)$$

where  $\mathcal{B}$  is a base topological space and  $\mathcal{M}_\phi$  is a smooth Riemannian manifold called the *phase target manifold*. Crucially,  $\mathcal{B}$  is not assumed to be a manifold, is not assumed to carry a metric, and is not assumed to have the dimension of spacetime. The topological structure of  $\mathcal{B}$  is itself emergent from the global organization of  $\Phi$ . Spacetime, as a four-dimensional Lorentzian manifold, is a derived object: it appears as the effective relational structure among certain classes of admissible configurations in appropriate limits. Similarly, matter fields, gauge fields, and geometry are all emergent from  $\Phi$ —they are not primitive ingredients of the theory but patterns encoded in the configuration of the phase map. Paper 1 argued that this primitive is motivated by the universality of phase in physical phenomena: the geometric phase in quantum mechanics [2, 3], the gauge phase in field theory [4], and the holonomy phase in gravity [5] all point toward phase as a more

fundamental quantity than the fields or spacetime structures over which it is defined.

The *phase configuration space*  $\mathfrak{C}$  is defined as the set of all maps  $(\Phi: \mathcal{B} \rightarrow \mathcal{M})_\Phi$  satisfying a minimal regularity condition, called Regularity Condition R, which specifies that  $(\Phi)$  is locally square-integrable with locally square-integrable first variations. This condition is the weakest condition under which the coherence and gradient structures of  $(\Phi)$  are well-defined as mathematical objects. The configuration space  $\mathfrak{C}$  is therefore a function space of considerable generality—it contains smooth maps, Sobolev-class maps, maps with isolated singularities, and maps with topologically nontrivial winding. It does not contain maps that are genuinely discontinuous on open sets, as such maps fail Regularity Condition R. The physical content of Phase Theory is the identification of a structured subset  $\mathfrak{C}_\mathcal{A} \subset \mathfrak{C}$  of *admissible configurations*—those that correspond to physical reality—and the derivation of all dynamical and structural properties of physics from this subset and its intrinsic geometry.

Paper 1 also introduced the notion of *emergent spacetime* within Phase Theory. The causal and metric structure of spacetime is not posited as a background but arises from the relational properties of admissible phase configurations. Roughly speaking, two regions of  $\mathcal{B}$  are causally connected if there exists an admissible evolution path connecting phase configurations supported on those regions. The metric emerges from the coherence structure of  $(\Phi)$ , as will be made precise in Section 9 and elaborated in Paper 8. The present paper takes all of these foundational results as given and focuses on the construction and analysis of the admissibility functional  $\mathcal{I}[\Phi]$ , which is the mechanism by which  $\mathfrak{C}_\mathcal{A}$  is identified within  $\mathfrak{C}$ .

### 3. The Necessity of Admissibility

The configuration space  $\mathfrak{C}$  is immense. It contains configurations corresponding to smoothly varying, coherent, particle-rich universes like our own, but it also contains configurations corresponding to maximally disordered, incoherent, featureless phase noise. If all elements of  $\mathfrak{C}$  were equally realizable—if no selection criterion operated—then the expected state of the universe would be one of maximal disorder, with no persistent structure of any kind. Yet persistent macroscopic structure is the most elementary empirical fact about the physical world: particles exist and are stable over timescales many orders of magnitude longer than the Planck time; atoms, molecules, and larger structures persist; spacetime geometry is smooth and regular on cosmological scales. This persistence is radically atypical with respect to the uniform measure on  $\mathfrak{C}$ . The inference is unavoidable: a strong, nonperturbative selection mechanism constrains the realized configurations to a tiny, highly structured subset of  $\mathfrak{C}$ . Any physical theory founded on a configuration space must either account for this selection or fail to explain the existence of persistent structure. The present section argues that the admissibility functional  $\mathcal{I}[\Phi]$  is the minimal object capable of implementing this selection within Phase Theory's ontological framework.

Define *structural persistence* as follows: a configuration  $\Phi \in \mathfrak{C}$  is structurally persistent if there exists a neighborhood  $\mathcal{U} \subset \mathfrak{C}$  of  $\Phi$  such that all configurations in  $\mathcal{U}$  share the topological and coherence properties of  $\Phi$  to within a specified tolerance. Persistence is therefore a property of a configuration relative to its neighborhood in configuration space, not a property of its temporal behavior—since time is not yet defined at this stage of the theory. Persistent configurations are those that are not isolated points in  $\mathfrak{C}$  but centers of structurally coherent regions. The selection mechanism required by observation must preferentially realize

persistent configurations over non-persistent ones. Any such mechanism can be characterized, in full generality, as a functional  $\mathcal{I}: \mathfrak{C} \rightarrow \mathbb{R}$  whose sign determines realizability. The simplest and most parsimonious such functional—compatible with the axioms enumerated in Section 4—is the admissibility functional introduced in this paper.

A central question is whether the selection mechanism must be intrinsic to  $(\Phi)$  or may be exogenous. Any exogenous selection rule introduces a second primitive alongside phase: the rule itself must be specified, and its specification constitutes an additional ontological commitment. Phase Theory, committed to phase as the sole primitive, cannot tolerate such exogeneity. The selection must therefore derive from properties of  $(\Phi)$  alone. This is not a mere aesthetic preference but a logical requirement: if selection could not be derived from  $(\Phi)$ , Phase Theory would not constitute a self-contained theory—it would require supplementation. The admissibility functional  $\mathcal{I}[\Phi]$ , defined as a function of  $(\Phi)$  via the axioms of Section 4, satisfies this requirement: it is fully determined by the global properties of the phase map, with no external input.

It is instructive to contrast this approach with the selection mechanisms in extant physical theories. In standard quantum mechanics, selection is probabilistic: all configurations exist in superposition, and the Born rule assigns probabilities to outcomes upon measurement. The Born rule is itself an exogenous postulate—it is not derived from the Schrödinger equation or the Hilbert space structure. In classical field theory, selection operates through equations of motion: physical trajectories are those satisfying the Euler-Lagrange equations of an action functional, which is again an exogenous postulate. In quantum gravity frameworks, selection criteria are imposed through path integral weights, spin foam amplitudes, or causal structure constraints, all of which require postulated structures beyond the



primitive. Phase Theory's admissibility approach is unique in that it operates *prior to probability*: admissibility determines which configurations can in principle exist, before any statistical question arises. Probability, as shown in Section 12, emerges as a derived concept from the admissibility structure when observational access is coarse-grained. This logical ordering—ontology first, statistics second—resolves the foundational incoherence of the Born rule's exogenous status in quantum mechanics.

## 4. Formal Definition of the Admissibility Functional

This section provides the precise mathematical definition of the admissibility functional and its governing axioms. Throughout,  $\mathcal{B}$  denotes a connected topological space and  $\mathcal{M}_\phi$  denotes a smooth, compact, connected Riemannian manifold without boundary.

### Definition 2.1 (Phase Configuration Space).

Let  $\mathcal{B}$  be a connected topological space and  $\mathcal{M}_\phi$  a smooth Riemannian manifold. The

phase configuration space

is

$$\mathcal{C} = \{ \Phi : \mathcal{B} \rightarrow \mathcal{M}_\phi \mid \Phi \text{ satisfies Regularity Condition R} \}$$

where

Regularity Condition R

specifies that  $\Phi$  is locally square-integrable with locally square-integrable first variations—i.e.,  $\Phi \in W^{1,2}_{\text{loc}}(\mathcal{B}, \mathcal{M}_\phi)$  in the sense of Sobolev maps into a Riemannian target.

**Definition 2.2 (Admissibility Functional).**

The

admissibility functional

is a map

$$\mathcal{I}: \mathcal{F}(C) \rightarrow \mathbb{R}$$

satisfying Axioms A1 through A5. A configuration  $(\Phi \in \mathcal{F}(C))$  is called physically admissible

if and only if  $(\mathcal{I}[\Phi] \geq 0)$ . The

admissible configuration space

$$\mathcal{F}(C)_{\mathcal{I}} = \mathcal{I}^{-1}([0, \infty)).$$

The five axioms governing  $(\mathcal{I})$  are as follows.

**Axiom A1 (Globality).**

$(\mathcal{I}[\Phi])$  is not expressible as an integral of a local density alone; it depends irreducibly on global properties of  $(\Phi)$ . Formally, there exists no local density  $(\ell: \mathcal{M}_{\Phi} \times \mathcal{M}_{\Phi} \rightarrow \mathbb{R})$  such that  $(\mathcal{I}[\Phi] = \int_{\mathcal{B}} \ell(\Phi, \nabla \Phi) \, d\mu)$  for all  $(\Phi \in \mathcal{F}(C))$ . The functional necessarily incorporates global information—topological winding numbers, holonomies, and large-scale coherence measures—that cannot be captured pointwise.

**Axiom A2 (Non-negativity of Physical Configurations).**

$\mathcal{I}[\Phi] \geq 0$  if and only if  $(\Phi)$  is physically admissible. The zero locus  $\partial \mathcal{A} = \mathcal{I}^{-1}(\{0\})$  constitutes the *admissibility boundary*—the marginal set of configurations that are realizable but not robustly so. Configurations with  $\mathcal{I}[\Phi] < 0$  are ontologically nonexistent.

**Axiom A3 (Topological Sensitivity).**

$\mathcal{I}$  assigns strictly positive values to configurations with nontrivial topological winding numbers that cannot be continuously deformed to zero within  $\mathcal{C}$ . Precisely, if  $(\Phi)$  represents a nontrivial element of  $\pi_n(\mathcal{M}_\phi)$  for some  $(n \geq 1)$ , and if no continuous path in  $\mathcal{C}$  connects  $(\Phi)$  to a topologically trivial configuration while maintaining  $\mathcal{I} \geq 0$ , then  $\mathcal{I}[\Phi] > 0$ .

**Axiom A4 (Coherence Monotonicity).**

For any family of configurations  $\{\Phi_\lambda\}_{\lambda \geq 0}$  parameterized by a decoherence parameter  $(\lambda)$  measuring the degree of incoherent oscillatory fragmentation of the phase profile,  $\mathcal{I}[\Phi_\lambda]$  is monotonically non-increasing in  $(\lambda)$  for all  $(\lambda)$  exceeding a critical threshold  $(\lambda_c)$ . That is, beyond  $(\lambda_c)$ , increasing disorder in the phase profile strictly decreases admissibility.

**Axiom A5 (Scale Neutrality).**

$\mathcal{I}$  does not depend on any externally imposed length scale, time scale, or energy scale. All scales that appear in the physical predictions of Phase Theory emerge from the internal structure of admissible configurations—from the coherence properties and topological charges of  $\Phi$  itself—and not from any parameter introduced into the definition of  $\mathcal{I}$ .

**Remark 2.1.**

The functional  $\mathcal{I}$  satisfying Axioms A1–A5 is not unique. All functionals satisfying these axioms define the same admissibility partition  $(\mathcal{C}_\mathcal{A}, \mathcal{C} \setminus \mathcal{C}_\mathcal{A})$  up to monotone reparameterization of  $\mathbb{R}$ . The physical content of the theory is encoded entirely in this partition, not in the specific numerical values assigned by  $\mathcal{I}$ . Two functionals  $\mathcal{I}$  and  $\mathcal{I}'$  are physically equivalent if and only if  $\mathcal{I}[\Phi] \geq 0 \iff \mathcal{I}'[\Phi] \geq 0$  for all  $\Phi \in \mathcal{C}$ .

**Remark 2.2.**

The condition  $\mathcal{I}[\Phi] < 0$  must be interpreted with care. It does not mean that the configuration  $\Phi$  is dynamically unstable, energetically costly, or improbable. It means that  $\Phi$  has no ontological status whatsoever—it does not exist in any physical sense, cannot be approximated by a sequence of physical configurations, and cannot appear as an intermediate state in any physical process. This is a strictly stronger claim

than dynamical suppression or probabilistic suppression, as elaborated in Section 10.

## 5. Canonical Decomposition of the Functional

While the admissibility functional  $\mathcal{I}[\Phi]$  is a unitary global object, it admits a conceptually illuminating canonical decomposition into three structurally distinct contributions:

$$\mathcal{I}[\Phi] = \mathcal{C}[\Phi] + \mathcal{T}[\Phi] + \mathcal{S}[\Phi]$$

This decomposition is structural rather than literally additive in the classical functional sense: it identifies three independent mechanisms through which  $\mathcal{I}$  enforces admissibility. The coherence term  $\mathcal{C}[\Phi]$  enforces phase regularity; the topological term  $\mathcal{T}[\Phi]$  rewards topologically nontrivial structure; and the stability term  $\mathcal{S}[\Phi]$  ensures robustness under perturbation. Each term is analyzed in turn.

### 5.1 Coherence Term $\mathcal{C}[\Phi]$

The coherence term  $\mathcal{C}[\Phi]$  penalizes excessive local phase gradients and incoherent oscillatory fragmentation of the phase profile. Let  $\nabla_B \Phi$  denote the generalized gradient of  $\Phi$  on  $B$ —the distributional derivative of the phase map in the sense of Sobolev maps into a Riemannian target. Define the coherence threshold  $\kappa_c > 0$  as the value of the  $(L^2)$ -norm of  $\nabla_B \Phi$  above which the phase profile is considered incoherent. Then  $\mathcal{C}[\Phi]$  is a decreasing functional of  $\|\nabla_B \Phi\|_{L^2}$  in the regime  $\|\nabla_B \Phi\|_{L^2} > \kappa_c$ , and is positive and bounded above in the coherent regime  $\|\nabla_B \Phi\|_{L^2} \leq \kappa_c$ . Configurations with bounded, smoothly varying

phase profiles—in which  $\Phi$  changes gradually across  $\mathcal{B}$ —contribute positively to  $\mathcal{C}$ . Highly fragmented or discontinuously varying configurations, except where topological protection operates (see §5.2), drive  $\mathcal{C}[\Phi]$  to large negative values, thereby rendering the configuration inadmissible via Axiom A2.

The coherence term has an important physical interpretation: it is the mechanism by which Phase Theory enforces the regularity of physical fields without any ultraviolet regulator. Propagating phase waves—coherent, smoothly varying disturbances in  $\Phi$ —satisfy  $|\nabla_{\mathcal{B}} \Phi|_{L^2} \leq \kappa_c$  and therefore contribute positively to  $\mathcal{C}$ . These propagating disturbances will be identified with photons in Paper 4, where the dispersion relation and polarization structure of admissible wave configurations are derived from the coherence constraint. The coherence term is also sensitive to the topology of  $\mathcal{B}$ : on simply connected regions, coherence is a purely local gradient condition; on non-simply connected regions, coherence additionally depends on the holonomy of  $\Phi$  around non-contractible loops, providing a natural coupling between phase coherence and the topology of the underlying space.

## 5.2 Topological Term $\mathcal{T}[\Phi]$

The topological term  $\mathcal{T}[\Phi]$  rewards topologically nontrivial configurations—those that cannot be continuously deformed to the topologically trivial map within  $\mathcal{C}$ . Let  $\pi_n(\mathcal{M}_{\Phi})$  denote the  $n$ -th homotopy group of the phase target manifold, where  $n$  is determined by the dimension of the effective support structure of  $\Phi$ . A configuration  $\Phi$  carrying winding number  $w \in \pi_n(\mathcal{M}_{\Phi}) \setminus \{0\}$  contributes positively to  $\mathcal{T}[\Phi]$ , with the contribution growing monotonically with the absolute value of the topological charge  $|w|$ . Topologically trivial configurations contribute zero to  $\mathcal{T}$ . This asymmetry—positive contribution from

topological structure, zero from topological triviality—implements the physical intuition that persistent particle-like structures are topologically protected: their existence is rewarded, not merely tolerated, by the admissibility functional.

The topological term is responsible for several key phenomena across the Phase Theory series. It accounts for the stability of particle-like defects—localized, topologically nontrivial excitations of  $\Phi$  that will be identified with elementary particles in Paper 3. It explains the quantization of conserved charges, which arise as the discrete elements of  $\pi_n(\mathcal{M}_\Phi)$  in Paper 7. It establishes the impossibility of continuous interpolation between distinct topological sectors: since any path in  $\mathcal{C}$  connecting configurations of different topological charge must pass through a configuration with discontinuous phase (which has  $\mathcal{C} \rightarrow -\infty$ ) and hence  $(\mathcal{I} < 0)$ , such interpolations are not physical evolutions. The topological term implements a form of topological protection: configurations that would require discontinuous deformation to destroy are permanently admissible, provided their coherence and stability contributions remain non-negative.

### 5.3 Stability Term $\mathcal{S}[\Phi]$

The stability term  $\mathcal{S}[\Phi]$  ensures that the admissibility of a configuration is not an artifact of fine-tuning—that small perturbations do not catastrophically alter the configuration's ontological status. Formally, define the *stability margin* of  $\Phi$  as

$$\sigma[\Phi] = \inf_{\{\|\delta\Phi\|_{L^2} < \epsilon\}} \big( \mathcal{I}[\Phi + \delta\Phi] - \mathcal{I}[\Phi] \big)$$

for a fixed  $\epsilon > 0$  small relative to the coherence scale. A configuration is called *robustly admissible* if  $\mathcal{I}[\Phi] \geq 0$  and  $\sigma[\Phi] > 0$ ; it is *dynamically marginal* if  $\mathcal{I}[\Phi] \geq 0$

and  $\sigma[\Phi] = 0$ ); and it is *spuriously admissible* if  $\mathcal{I}[\Phi] \geq 0$  but  $\sigma[\Phi] < 0$ . Spuriously admissible configurations are those that, while formally satisfying  $\mathcal{I} \geq 0$ , are immediately rendered inadmissible by arbitrarily small perturbations. Such configurations do not persist in physical reality: the stability term  $\mathcal{S}[\Phi]$  is defined so that spuriously admissible configurations have  $\mathcal{S}[\Phi] < 0$  of sufficient magnitude to drive the full functional negative under perturbation. Robustly admissible configurations, by contrast, have  $\mathcal{S}[\Phi] > 0$ , contributing positive support against perturbative deformation out of the admissible set. It is robustly admissible configurations that correspond to the genuinely physical structures of Phase Theory.

## 6. The Coherence Functional: Detailed Structure

The coherence term  $\mathcal{C}[\Phi]$  admits a more detailed analysis than the schematic description of §5.1. A central concept is the *phase coherence length*  $\ell_c(\Phi)$ , defined as the characteristic scale over which  $\Phi$  varies by order unity in the metric of  $\mathcal{M}_\Phi$ . More precisely,  $\ell_c(\Phi)$  is the largest value of  $r$  such that for almost every point  $x \in \mathcal{B}$ , the oscillation of  $\Phi$  in the  $r$ -neighborhood of  $x$  (measured in any natural metric on  $\mathcal{B}$  compatible with its topology) is less than a reference variation  $\Delta_0$  determined by  $\mathcal{M}_\Phi$ . The coherence term  $\mathcal{C}[\Phi]$  is an increasing functional of  $\ell_c(\Phi)$ : configurations with large coherence length—slowly, smoothly varying phase profiles—contribute positively, while configurations with short coherence length—rapidly varying, fragmented profiles—contribute negatively or not at all.

The dependence of coherence on the topology of  $\mathcal{B}$  is subtle. On simply connected regions of  $\mathcal{B}$ , coherence is indeed a purely local gradient condition:  $\Phi$  is coherent if and only if  $\|\nabla \mathcal{B} \Phi\|_{L^\infty}$  is bounded above by the coherence threshold.



On non-simply connected regions—those with nontrivial fundamental group  $\pi_1(\mathcal{B})$ —the coherence of  $(\Phi)$  additionally depends on the holonomy of the phase map around non-contractible loops  $(\gamma \subset \mathcal{B})$ . Specifically, the holonomy  $(\text{Hol}_\gamma(\Phi) = \oint_\gamma d\Phi)$  contributes to the gradient norm and hence to the coherence assessment. This holonomy dependence provides the mechanism by which Phase Theory generates gauge-like structures from the coherence requirement alone: the admissibility functional naturally penalizes configurations whose holonomies are inconsistent with global coherence, reproducing the role of gauge field curvature in standard physics (Paper 7).

The coherence structure of  $(\mathcal{C}[\Phi])$  implies a fundamental lower bound on the coherence length of any admissible configuration. This is formalized in the following lemma.

**Lemma 6.1 (Coherence Bound).**

For any  $(\Phi \in \text{mathfrak{C}})$  with  $(\mathcal{I}[\Phi] \geq 0)$ , the coherence length satisfies

$$(\ell_c(\Phi) > \ell_{\min})$$

where  $(\ell_{\min})$

>

0) is a universal lower bound determined solely by the topology and Riemannian geometry of  $(\mathcal{M}_\phi)$ , independent of any externally imposed scale.

**Proof Sketch.**

Suppose for contradiction that  $(\ell_c(\Phi) \leq \ell_{\min})$ . By definition of the coherence length, this implies that  $(\Phi)$  varies by order unity over scales of size  $(\ell_c(\Phi))$ .

$(\ell_{\min})$  or smaller. Consequently, the  $(L^2)$ -norm of  $(\nabla_{\mathcal{B}} \Phi)$  exceeds the coherence threshold  $(\kappa_c)$ , which by the definition of  $(\mathcal{C}[\Phi])$  implies  $(\mathcal{C}[\Phi] < 0)$ . For  $(\mathcal{I}[\Phi] \geq 0)$  to hold despite  $(\mathcal{C}[\Phi] < 0)$ , the topological and stability terms must compensate. However, by Axiom A5 (Scale Neutrality) and the compactness of  $(\mathcal{M}_\phi)$ , the topological contribution  $(\mathcal{T}[\Phi])$  is bounded above by a constant  $(T_{\max})$  determined by  $(\pi_n(\mathcal{M}_\phi))$ , and the stability term  $(\mathcal{S}[\Phi])$  is bounded above by  $(S_{\max})$  similarly determined by the geometry of  $(\mathcal{M}_\phi)$ . As  $(\ell_c \rightarrow 0)$ ,  $(\mathcal{C}[\Phi] \rightarrow -\infty)$ , so for sufficiently small  $(\ell_c)$  the combined contribution  $(\mathcal{T} + \mathcal{S} \leq T_{\max} + S_{\max})$  cannot compensate, and  $(\mathcal{I}[\Phi] < 0)$ . The threshold  $(\ell_{\min})$  is the infimum of coherence lengths for which this compensation is marginally possible, and its value is determined by the geometry of  $(\mathcal{M}_\phi)$  via  $(\ell_{\min} = (T_{\max} + S_{\max})^{-1/2})$  in appropriate units.

□

The coherence bound of Lemma 6.1 has profound physical implications. It establishes a natural minimum scale below which phase configurations cannot be coherent and hence cannot be admissible: Phase Theory contains an intrinsic ultraviolet regularization that is a consequence of admissibility rather than a postulated cutoff. This is in sharp contrast to standard quantum field theory, which requires renormalization precisely because the theory contains no intrinsic mechanism for excluding arbitrarily short-distance field configurations. In Phase Theory, arbitrarily short-wavelength configurations are not suppressed by renormalization group running—they are ontologically nonexistent. This distinction has concrete implications for the ultraviolet behavior of scattering amplitudes derived from Phase Theory (Paper 5), which are expected to be finite without renormalization for observables computed strictly within the admissible configuration space.

A further implication of Lemma 6.1 is the emergence of a natural length scale  $\ell_{\min}$  from the geometry of  $(\mathcal{M}, \phi)$  alone. This scale plays the role of the Planck length in Phase Theory—the minimum scale below which the theory predicts no physical content—but its precise value is derivable from the structure of the phase manifold rather than postulated by dimensional analysis. Determining the explicit value of  $\ell_{\min}$  in terms of known physical constants requires the identification of  $(\mathcal{M}, \phi)$ , which is addressed in Paper 3. The existence of  $\ell_{\min}$  as a derived rather than postulated quantity illustrates the explanatory power of the admissibility framework: what appears in conventional physics as a fundamental, unexplained scale emerges here as a consequence of the geometry of the phase target manifold.

## 7. Topological Sectors and Protected Configurations

The configuration space  $\mathfrak{C}$  admits a natural decomposition into topological sectors, defined as equivalence classes of configurations under continuous deformation within  $\mathfrak{C}$ . Two configurations  $\Phi_1, \Phi_2 \in \mathfrak{C}$  are in the same topological sector if and only if there exists a continuous path  $\gamma: [0,1] \rightarrow \mathfrak{C}$  with  $\gamma(0) = \Phi_1$  and  $\gamma(1) = \Phi_2$ . The quotient of  $\mathfrak{C}$  by this equivalence relation is the set of topological sectors, indexed—under appropriate conditions on  $(\mathcal{B})$  and  $(\mathcal{M}, \phi)$ —by the homotopy group  $\pi_n(\mathcal{M}, \phi)$ . The topological term  $\mathcal{T}[\Phi]$  is constant within each sector and jumps discontinuously across sector boundaries, reflecting the fact that the topological charge is an integer-valued invariant.

### Definition 7.1 (Topological Charge).

For  $\Phi \in \mathfrak{C}$ , the *topological charge*  $Q(\Phi) \in \pi_n(\mathcal{M}, \phi)$

$\mathcal{M}_\Phi$  is the homotopy class of  $\Phi$  as a map from (the appropriate compactification of)  $\mathcal{B}$  into  $\mathcal{M}_\Phi$ . This class is well-defined under Regularity Condition R and is invariant under continuous deformations within  $\mathcal{C}$  that do not change the boundary conditions of  $\Phi$ .

### Theorem 7.1 (Topological Protection).

Let  $\Phi \in \mathcal{C}_\mathcal{A}$  be a robustly admissible configuration with  $Q(\Phi) \neq 0$  in  $\pi_n(\mathcal{M}_\Phi)$ . Then no continuous deformation of  $\Phi$  within  $\mathcal{C}$  can reduce  $Q(\Phi)$  to zero while maintaining  $\mathcal{I}[\cdot] \geq 0$  throughout the deformation.

### Proof.

Let  $\gamma: [0,1] \rightarrow \mathcal{C}$  be any continuous path with  $\gamma(0) = \Phi$  and  $Q(\gamma(1)) = 0$ . Since  $Q$  is an integer-valued homotopy invariant, it cannot change continuously; it must jump at some parameter value  $t^* \in (0,1]$ . At such a jump, the map  $\gamma(t^*)$  is not homotopic to  $\gamma(t^* - \varepsilon)$  for any  $\varepsilon > 0$ , which means  $\gamma(t^*)$  cannot be obtained from  $\gamma(t^* - \varepsilon)$  by a pointwise continuous deformation. This requires the existence of a point (or set of points) in  $\mathcal{B}$  at which  $\gamma(t^*)$  is discontinuous. By Regularity Condition R, such a configuration fails the Sobolev regularity requirement and hence is not in  $\mathcal{C}$ . Alternatively, if  $\gamma(t^*)$  is approached as a limit within  $\mathcal{C}$ , the limiting configuration has a gradient norm that diverges—specifically,  $\|\nabla_\mathcal{B} \gamma(t^*)\|_{L^2} \rightarrow \infty$ —which drives  $\mathcal{C}[\gamma(t^*)] \rightarrow -\infty$  and hence  $\mathcal{I}[\gamma(t^*)] \rightarrow -\infty$ . Therefore,  $\gamma(t^*)$  is inadmissible, and the deformation necessarily exits

the admissible set. Since  $\mathcal{I}[\gamma(t^*)] < 0$ , the condition  $\mathcal{I}[\cdot] \geq 0$  is violated throughout any such deformation at  $(t^*)$ .

□

### Corollary 7.1.

Topologically charged admissible configurations are *absolutely stable* within Phase Theory: they cannot decay, annihilate, or otherwise lose their topological character without crossing an inadmissible configuration, which is ontologically nonexistent. No physical process can cause their topological charge to change.

Corollary 7.1 has immediate implications for the theory of elementary particles, developed in Paper 3. The stable elementary particles of Phase Theory are precisely the robustly admissible configurations with nontrivial topological charge. Their stability is not explained by a conservation law postulated independently—it follows necessarily from the structure of the admissibility functional and Theorem 7.1. A particle is stable not because a symmetry protects it but because its destruction would require ontologically nonexistent intermediate configurations. This is a stronger protection mechanism than any dynamical conservation law, since conservation laws can be broken by symmetry-violating interactions, whereas ontological elimination cannot be circumvented by any physical process.

The discrete structure of the group  $(\pi_n(\mathcal{M}_\phi))$  directly implies the quantization of topological charge. Since  $(\pi_n(\mathcal{M}_\phi))$  is a discrete (typically finitely generated abelian) group, topological charges take values in a discrete set. This provides the mechanism by which charge quantization arises in Phase Theory without any Dirac quantization

condition or other external argument: quantization is a property of the homotopy group of the phase manifold. The specific groups  $\pi_n(\mathcal{M}_\phi)$  for the particular  $\mathcal{M}_\phi$  proposed in Paper 3 yield electric charge quantization in units of  $(e/3)$ , angular momentum quantization in units of  $(\hbar/2)$ , and baryon number quantization in units of 1, recovering the observed charge structure of the Standard Model. A detailed derivation is given in Paper 7.

The connection between topological sectors and conserved quantities is deepened by the following observation. Within each sector  $\mathcal{C}_Q$ , the topological charge  $(Q)$  is a conserved label. The partition  $\mathcal{C} = \bigsqcup_Q \mathcal{C}_Q$  (established rigorously in Section 18) means that physical evolution within a sector cannot produce configurations in a different sector—a statement that is precisely analogous to the conservation of a quantum number but derived from topology rather than Noether's theorem. The relationship between topological conservation and Noether conservation is elaborated in Paper 7: the two approaches are shown to coincide in the smooth coherent limit, with topological conservation being the more fundamental, non-perturbative statement.

## 8. Admissible Evolution: Formal Definition and Properties

The notion of *physical evolution* in Phase Theory is formalized without reference to time, which is itself a derived concept (Paper 9). Evolution is defined as a relation on the admissible configuration space  $\mathcal{C}_\mathcal{A}$ , specifying which transitions between configurations are physically possible.

**Definition 8.1 (Admissible Path).**

A path  $\gamma: [0,1] \rightarrow \mathcal{C}$  is an

admissible evolution path

if the following three conditions hold:

1.  $I[\gamma(t)] \geq 0$  for all  $t \in [0,1]$ ;
2.  $I[\gamma(t)]$  does not decrease by more than a critical margin  $\Delta_c > 0$  over any sub-interval of measure less than  $\epsilon_0 > 0$ —that is, admissibility is not merely satisfied at every point but is maintained with a uniform lower bound against abrupt loss;
3. The topological charge  $Q(\gamma(t))$  is piecewise constant in  $t$ —topological sector transitions do not occur during physical evolution.

### Definition 8.2 (Physical Evolution).

A transformation  $\Phi_1 \mapsto \Phi_2$  is *physical* if and only if there exists an admissible evolution path  $\gamma: [0,1] \rightarrow \mathcal{C}$  with  $\gamma(0) = \Phi_1$  and  $\gamma(1) = \Phi_2$ . The set of all physically accessible configurations from  $\Phi_1$  is the *future orbit* of  $\Phi_1$ , denoted  $\mathcal{O}^+(\Phi_1)$ .

### Remark 8.1.

There is no fundamental notion of time in Definitions 8.1 and 8.2. The parameter  $t \in [0,1]$  is a formal ordering parameter for the path—a mathematical convenience—not a physical time coordinate. The emergence of

time as an ordered relational structure among admissible configurations, arising from the asymmetry of admissible evolution paths (the partial order of Theorem 8.1), is derived in Paper 9. All temporal language used heuristically in earlier sections (e.g., "evolution," "stability") refers to this emergent relational order, not to any background time parameter.

### Theorem 8.1 (Transitivity of Physical Evolution).

If  $(\Phi_1 \mapsto \Phi_2)$  and  $(\Phi_2 \mapsto \Phi_3)$  are physical evolutions in the sense of Definition 8.2, then  $(\Phi_1 \mapsto \Phi_3)$  is a physical evolution.

#### Proof.

Let  $(\gamma_1: [0,1] \rightarrow \mathcal{C})$  be an admissible path from  $(\Phi_1)$  to  $(\Phi_2)$ , and  $(\gamma_2: [0,1] \rightarrow \mathcal{C})$  an admissible path from  $(\Phi_2)$  to  $(\Phi_3)$ . Define the concatenated path  $(\gamma: [0,1] \rightarrow \mathcal{C})$  by  $(\gamma(t) = \gamma_1(2t))$  for  $(t \in [0, 1/2])$  and  $(\gamma(t) = \gamma_2(2t-1))$  for  $(t \in [1/2, 1])$ . By construction,  $(\gamma(0) = \Phi_1)$  and  $(\gamma(1) = \Phi_3)$ . Condition (i) of Definition 8.1 holds for  $(\gamma)$  since it holds for both  $(\gamma_1)$  and  $(\gamma_2)$ . Condition (ii) holds since both paths satisfy the margin condition independently, and the concatenation at  $(t = 1/2)$  is smooth because  $(\gamma_1(1) = \gamma_2(0) = \Phi_2)$ . Condition (iii) holds since topological charge is piecewise constant on both paths and hence on the concatenation. Therefore  $(\gamma)$  is an admissible evolution path and  $(\Phi_1 \mapsto \Phi_3)$  is physical.

□



Theorem 8.1, combined with the reflexivity ( $(\Phi \mapsto \Phi)$  is trivially physical) and the definition of the partial order relation induced by admissible evolution, equips the admissible configuration space  $(\mathfrak{C} \setminus \mathcal{A})$  with the structure of a preordered set. This preorder is the mathematical precursor of the temporal order that emerges in Paper 9. The partial order is strict (antisymmetric) if and only if the admissible evolution relation is not time-reversal invariant—that is, if  $(\Phi_1 \mapsto \Phi_2)$  does not imply  $(\Phi_2 \mapsto \Phi_1)$  in general. The time-asymmetry of the admissible evolution relation, which underlies the thermodynamic arrow of time, is governed by the stability term  $(\mathcal{S}[\Phi])$  and is analyzed in Paper 9.

## 9. Emergence of the Action Principle

One of the most consequential results of Phase Theory is the derivation of the action principle from the admissibility condition, rather than its postulation as a fundamental axiom. This section establishes the precise regime and sense in which extremization of an effective action emerges from the global admissibility condition  $(\mathcal{I}[\Phi] \geq 0)$ .

Let  $(\Phi_\lambda)$  be a smooth one-parameter family of configurations parameterized by  $(\lambda \in \mathbb{R})$ , with  $(\Phi_0 = \Phi)$  the configuration of interest. Define the Fréchet derivative of  $(\mathcal{I})$  along this family as

$$(\delta_\lambda \mathcal{I} = \frac{d}{d\lambda} \mathcal{I}[\Phi_\lambda] \Big|_{\lambda=0})$$

The following theorem establishes that robust admissibility implies first-order stationarity of  $(\mathcal{I})$  in smooth regimes.

### Theorem 9.1 (Emergent Extremization).

Suppose that (i)  $(\mathcal{B})$  carries a smooth manifold structure; (ii)  $(\mathcal{I})$

$\mathcal{M}_\phi$ ) is a compact Riemannian manifold; (iii)  $(\Phi \in \mathfrak{C}_\mathcal{A})$  is robustly admissible with  $(\sigma[\Phi]$

>

0); and (iv) phase variations  $(\delta\Phi)$  are small relative to the coherence scale:  $(\|\delta\Phi\|_{L^\infty} \ll \ell_c(\Phi)^{-1})$ . Then to leading order in the variation,  $(\delta_\lambda \mathcal{I}[\Phi] = 0)$

for all smooth, compactly supported variations  $(\delta\Phi)$  preserving the topological sector of  $(\Phi)$ .

### Proof Sketch.

Under the stated conditions,  $(\mathcal{I}[\Phi])$  is Fréchet differentiable at  $(\Phi)$  as a functional on  $(\mathfrak{C})$  (equipped with the  $(L^2)$  norm topology), which follows from the smoothness of  $(\mathcal{B})$ , the compactness of  $(\mathcal{M}_\phi)$ , and the regularity of  $(\Phi)$ . Since  $(\Phi)$  is robustly admissible with  $(\sigma[\Phi] > 0)$ , the configuration  $(\Phi)$  is a local maximizer of  $(\mathcal{I})$  within its topological sector  $(\mathfrak{C}_Q)$ : any small perturbation preserving topological charge either leaves  $(\mathcal{I})$  unchanged to first order or increases it, with the bound on increase governed by  $(\sigma[\Phi])$ . For a local interior maximum of a Fréchet-differentiable functional, the first-order variation vanishes:  $(\delta_\lambda \mathcal{I}|_{\lambda=0} = 0)$ . This is the statement that  $(\Phi)$  is a stationary point of  $(\mathcal{I})$  within its sector, which is precisely the extremization condition.

□

### Corollary 9.1.

Classical action principles are effective approximations to the admissibility condition, valid in smooth, coherent, small-variation regimes. They are not fundamental postulates of physical law but emergent descriptions.

The Euler-Lagrange equations of classical and quantum field theories emerge as the first-order conditions of the extremization stated in Theorem 9.1. Specifically, the stationarity condition  $\delta \mathcal{I} = 0$  translates, via the Fréchet derivative of  $\mathcal{I}$ , into a set of second-order partial differential equations for  $\Phi$  on  $\mathcal{B}$ . In the smooth coherent regime, these equations take the schematic form of standard field equations. The precise identification of the emergent field equations with the Standard Model equations of motion and the Einstein field equations is carried out in Papers 7 and 8. Higher-order corrections to the emergent extremization—arising from the non-linearity of  $\mathcal{I}$ —beyond the smooth coherent regime—correspond to loop corrections in quantum field theory (Paper 5), establishing a precise connection between the admissibility functional and the quantum effective action.

In the smooth coherent regime, the effective action takes the form

$$\mathcal{S}_{\text{eff}}[\Phi] = \int_{\mathcal{B}} \mathcal{L}_{\text{eff}}(\Phi, \nabla \Phi) \, d\mu_{\mathcal{B}}$$

where  $\mathcal{L}_{\text{eff}}$  is an effective Lagrangian density determined by the local properties of  $\mathcal{I}$ , and  $d\mu_{\mathcal{B}}$  is an emergent measure on  $\mathcal{B}$  derived from the phase structure—not assumed as a background volume form. The measure  $d\mu_{\mathcal{B}}$  plays the role of  $\sqrt{-g} \, d^4x$  in general relativity, but it is not postulated; it arises from the density of admissible phase configurations on  $\mathcal{B}$ . This derivation is the key step through which general covariance emerges in Phase Theory: the effective action is

automatically generally covariant because the measure is derived from  $\int \Phi$  rather than a fixed background metric.

It is important to emphasize the logical direction of this emergence. In standard physics, one begins with an action functional—postulated—and derives equations of motion by varying it. In Phase Theory, one begins with the admissibility condition—derived from the axioms of Phase Theory—and extracts an effective action as the leading-order description of that condition in smooth coherent regimes. The action is therefore not an independent postulate but a compressed encoding of the admissibility condition's first-order behavior. Its limitations—the need for renormalization, the emergence of non-perturbative effects, the Haag-Kastler axiom violations—all correspond to regions where the smooth coherent approximation breaks down and the full admissibility condition must be used. The admissibility functional is thus the fundamental object of which the action is an approximation, not the other way around.

## 10. Forbidden Configurations and Ontological Elimination

The central conceptual innovation of Phase Theory—and its sharpest departure from all conventional frameworks—is the ontological status of inadmissible configurations. Configurations with  $\int \Phi < 0$  are not merely improbable, energetically disfavored, dynamically suppressed, or probabilistically negligible. They do not exist. No physical observation, measurement, or process can make reference to them, because they have no physical reality whatsoever. This is the principle of *ontological elimination*, and it is the foundational consequence of Axiom A2.

To appreciate the force of this distinction, it is instructive to contrast it with the role of forbidden configurations in standard theories. In standard quantum mechanics, all configurations of the Hilbert space exist simultaneously in superposition; the Born rule assigns probability

amplitudes, but no configuration is ontologically absent—every Hilbert space vector is equally real as a potential state. The measurement process collapses this superposition, but even this collapse is often interpreted as an epistemic update rather than an ontological change. In classical field theory, all configurations of the field space exist as mathematical possibilities; the equations of motion select a trajectory through this space, and only the selected trajectory is "physical," but the others remain well-defined mathematical objects. In Path Integral quantum field theory, indeed, all field configurations are integrated over—each contributing a phase factor—with no configuration being absent from the sum. In each case, the selection of physical configurations from the space of all possibilities is a dynamical or statistical operation, not an ontological one. Phase Theory breaks with this tradition entirely: the inadmissible configurations are not merely unselected—they are nonexistent.

### Theorem 10.1 (Ontological Elimination).

Let  $(\Phi_f \in \mathfrak{C})$  be a configuration with  $(\mathcal{I}[\Phi_f]$

$<$

$0$ ). Then:

1.  $(\Phi_f)$  is not a member of any physical superposition of states;
2. No physical evolution path starting from any admissible  $(\Phi_i \in \mathfrak{C}_{\mathcal{A}})$  can reach  $(\Phi_f)$ ;
3. No observable quantity can be evaluated at  $(\Phi_f)$ .

### Proof.

(i) In Phase Theory, superpositions of phase configurations are defined, in the

sense relevant to Paper 5, over branches in the admissible configuration space  $(\mathcal{C} \setminus \mathcal{A})$ . The definition of physical superposition in Paper 5 explicitly restricts the domain of coherent branching to  $(\mathcal{C} \setminus \mathcal{A})$ . Since  $(\Phi_f \notin \mathcal{C} \setminus \mathcal{A})$ , it cannot appear as a branch. (ii) By Definition 8.1, every admissible evolution path  $(\gamma)$  satisfies  $(\mathcal{I}[\gamma(t)] \geq 0)$  for all  $(t \in [0,1])$ . Therefore  $(\gamma(1) \in \mathcal{C} \setminus \mathcal{A})$ , and no path reaching  $(\Phi_f \notin \mathcal{C} \setminus \mathcal{A})$  qualifies as admissible. (iii) Observables in Phase Theory are defined as functionals  $(\mathcal{O}: \mathcal{C} \setminus \mathcal{A} \rightarrow \mathbb{R})$  on the admissible configuration space. Their domain is  $(\mathcal{C} \setminus \mathcal{A})$  by definition, so they are not defined at  $(\Phi_f)$ .

□

The principle of ontological elimination has far-reaching consequences for the resolution of apparent paradoxes in quantum mechanics and quantum gravity. Many paradoxes—the Schrödinger's cat problem, the black hole information paradox, certain singularity theorems in general relativity—arise, on close inspection, from the implicit assignment of physical meaning to configurations that, within Phase Theory, are inadmissible. The superposition of macroscopically distinct configurations (dead and alive cat) is admissible only if both branches satisfy  $(\mathcal{I} \geq 0)$  throughout; configurations that would require macroscopic coherence over scales exceeding the coherence bound of Lemma 6.1 are inadmissible and hence cannot appear in any superposition. The black hole singularity corresponds to a configuration where the emergent spacetime geometry becomes singular—a configuration that, as argued in Paper 8, has  $(\mathcal{I}[\Phi] < 0)$  and is therefore ontologically absent from Phase Theory. These resolutions are not ad hoc; they follow necessarily from the structure of  $(\mathcal{I}[\Phi])$  without additional postulates or interpretation choices.

## 11. Discreteness as a Consequence of Admissibility

One of the deepest physical consequences of the admissibility functional is the emergence of discrete structure throughout Phase Theory—without any quantization postulate, lattice approximation, or discretization of spacetime. Discreteness arises as a mathematical consequence of the combined constraints imposed by the coherence bound (Lemma 6.1) and the requirement of robust admissibility (§5.3).

### Theorem 11.1 (Spectral Discreteness).

Let  $\{\Phi_n\}_{n \in \mathbb{N}}$  be the set of robustly admissible configurations within a topological sector  $[Q]$  of  $\mathcal{C}_A$ . Then the spectrum of any functional  $\mathcal{F}: \mathcal{C} \rightarrow \mathbb{R}$  that is (a) continuous with respect to the  $(L^2)$  topology on  $\mathcal{C}$ , (b) invariant under small perturbations (i.e.,  $|\mathcal{F}[\Phi + \delta\Phi] - \mathcal{F}[\Phi]| < \eta$  for  $\|\delta\Phi\|_{L^2} < \epsilon$ ), and (c) bounded above within  $[Q]$ , is a discrete subset of  $\mathbb{R}$  on the set  $\{\Phi_n\}$  of robustly admissible configurations.

### Proof Sketch.

By the coherence bound (Lemma 6.1), every robustly admissible configuration  $\Phi_n$  satisfies  $\ell_c(\Phi_n) > \ell_{\min}$ . This provides a lower bound on the spatial scale over which  $\Phi_n$  can vary. Combined with the requirement  $\sigma[\Phi_n] > 0$  (robust admissibility, §5.3), each  $\Phi_n$  is isolated within its topological sector: there exists an open  $(L^2)$ -ball around  $\Phi_n$  containing no other robustly admissible configuration. (This isolation follows from the fact that if two distinct robustly admissible configurations were arbitrarily close in  $(L^2)$ , their convex combination would be admissible with  $\sigma = 0$ ), contradicting

robust admissibility of both.) Therefore the set  $\{\Phi_n\}$  is a discrete subset of  $\left(\frac{C}{L^2}\right)$  in the  $(L^2)$  topology. Since  $\mathcal{F}$  is continuous and  $\{\Phi_n\}$  is discrete, the image  $\mathcal{F}(\{\Phi_n\})$  is a discrete subset of  $\mathbb{R}$ .

□

### Corollary 11.1.

Energy spectra, charge spectra, angular momentum spectra, and all other spectra of continuous observables are discrete within each topological sector of Phase Theory. This discreteness is not a postulate—it is a mathematical consequence of admissibility and the coherence bound.

The significance of this result cannot be overstated. Quantization—the discreteness of physical spectra—is typically introduced in quantum mechanics as a postulate, through canonical commutation relations or the Bohr-Sommerfeld quantization conditions, and in quantum field theory through the imposition of Fock space structure. In Phase Theory, quantization is a theorem: it follows from the geometry of the admissible configuration space. The discrete spectrum of allowed energy levels of a hydrogen atom, for example, corresponds to the discrete set of robustly admissible configurations in the relevant topological sector of  $\left(\frac{C}{L^2}\right)_A$ . The Rydberg formula emerges not from Bohr's ad hoc quantization rule but from the combinatorics of the admissible configuration set, as derived in Paper 3.

The contrast with the string theory landscape [6] is instructive. In string theory, the large number of consistent vacuum configurations—of order  $(10^{500})$ —is a problem: the theory predicts too many possibilities, with



no selection mechanism. The Phase Theory admissibility functional provides precisely such a mechanism: of the vast space  $\mathfrak{C}$ , only a discrete and highly constrained set satisfies all the conditions of Definition 2.2 with robust admissibility. The number of distinct topological sectors of  $\mathcal{M}_\Phi$  that support robustly admissible configurations is predicted to be approximately  $(25 \pm 10)$ , corresponding to the observed number of stable particle species (Paper 3). This dramatic reduction from the string landscape to a finite, predictive set is a direct consequence of the admissibility functional's stringent constraints.

A further implication of Theorem 11.1 concerns the continuum limit of physical quantities. In Phase Theory, there is no regime in which physical spectra become truly continuous—the discreteness imposed by Lemma 6.1 and robust admissibility is an exact feature, not an approximation. What appears as a continuous spectrum in, e.g., scattering cross-sections at high energies is a consequence of the high density of robustly admissible configurations in that regime, not of any genuine continuum. The apparent continuum of classical physics arises in the limit of large topological charge and long coherence length, where the spacing between discrete levels becomes negligible compared to the resolution of classical measurements. This is the sense in which the classical continuum is an emergent, approximate description within Phase Theory.

## 12. Relation to Probability and Statistical Behavior

The admissibility functional  $\mathcal{I}[\Phi]$  is a purely deterministic, non-probabilistic object. It assigns a definite real number to each phase configuration and partitions  $\mathfrak{C}$  into admissible and inadmissible sets without any recourse to probability theory. In particular, no probability measure is postulated as part of the axiomatic structure of Phase Theory: probability does not appear in Axioms A1–A5, in the definition of admissible evolution (Definition 8.1), or in the statement of any theorem of

the present paper. This is a deliberate and fundamental choice, reflecting the foundational conviction that probability is an emergent description of epistemic limitations, not a primitive feature of physical reality.

Statistical behavior emerges in Phase Theory only under two simultaneous conditions: (a) multiple distinct admissible configurations are compatible with the available boundary or initial data specifying the physical situation; and (b) observational access is coarse-grained—the observer cannot distinguish among the compatible admissible configurations with the available measurement apparatus. Under these conditions, an effective probability measure  $\mu_{\text{eff}}$  can be defined over the set of admissible configurations consistent with the coarse-grained data, and statistical predictions can be made by averaging over this measure. The specific form of  $\mu_{\text{eff}}$  depends on the structure of the admissible configuration space and the coarse-graining map, as derived in Paper 5.

### **Theorem 12.1 (Emergent Born Rule).**

In the limit of maximal coherence (large  $\ell_c$ ) and minimal coarse-graining (fine-grained observational access approaching the coherence scale), the effective probability measure  $\mu_{\text{eff}}$  over the admissible configurations consistent with given coarse-grained data converges to the squared-modulus measure of quantum mechanics:

$$\mu_{\text{eff}}(\Phi) \longrightarrow |\psi[\Phi]|^2$$

where  $\psi[\Phi]$  is the phase-theoretic amplitude functional defined in Paper 5.

Theorem 12.1 is stated here without full proof, as its derivation requires the amplitude formalism developed in Paper 5. The key conceptual point is the following: the Born rule is not an axiom of Phase Theory but a theorem—it

emerges, in the appropriate limit, from the geometry of the admissible configuration space and the coarse-graining structure of the measurement process. This resolves the long-standing foundational problem of the Born rule's exogenous status in quantum mechanics [7]. In standard quantum mechanics, the Born rule cannot be derived from the Schrödinger equation and must be postulated independently; its physical justification has been the subject of extensive debate [8, 9]. In Phase Theory, its emergence is a calculable consequence of more primitive assumptions.

The implications for the quantum measurement problem are significant. In standard quantum mechanics, the measurement problem asks why, given that the Schrödinger equation predicts a superposition of macroscopic outcomes after measurement, only one outcome is observed. Various interpretations—Copenhagen, many-worlds, decoherence, spontaneous collapse—address this problem with varying degrees of success but without consensus [7, 10]. In Phase Theory, definite outcomes arise because the admissibility constraint, applied to the measurement apparatus plus measured system as a combined phase configuration, selects a unique robustly admissible configuration as the outcome. The apparent superposition of outcomes is an artifact of treating the measurement process with the smooth-coherent effective action of Section 9, which ignores the admissibility structure. When the full  $\mathcal{I}[\Phi]$  is applied, only one branch survives as robustly admissible. No collapse postulate is needed: the collapse is implemented by the ontological elimination of all inadmissible branches. This resolution is elaborated in detail in Paper 6.

### **13. Comparison with Classical and Quantum Action Principles**

Phase Theory's admissibility functional represents a conceptually distinct approach to the selection of physical configurations from any previously

proposed framework. The following table summarizes the comparison across the principal frameworks of theoretical physics.

| Framework                                      | Foundational Object                                       | Requires Background Spacetime   | Probabilistic Foundation         | Trajectory-Based             | Dynamical Principle          |
|------------------------------------------------|-----------------------------------------------------------|---------------------------------|----------------------------------|------------------------------|------------------------------|
| Classical Mechanics (Least Action)             | Lagrangian $(L(q, \dot{q}, t))$                           | Yes (time background)           | No                               | Yes                          | Postulated                   |
| Classical Field Theory                         | Field action $(S[\phi] = \int_{\mathcal{L}} d^4x)$        | Yes (Minkowski or curved)       | No                               | Yes                          | Postulated                   |
| Quantum Mechanics (Path Integral)              | Path integral $(\int_{\mathcal{D}} \phi, e^{iS/\hbar})$   | Yes (background metric assumed) | Yes (Born rule postulated)       | No (all paths summed)        | Postulated                   |
| General Relativity (Einstein-Hilbert)          | Action $(S = \frac{1}{16\pi G} \int_R \sqrt{-g} d^4x)$    | Self-referential (dynamic)      | No                               | Yes (geodesics)              | Postulated                   |
| <b>Phase Theory (Admissibility Functional)</b> | $(\mathcal{I}[\Phi]: \mathcal{C} \rightarrow \mathbb{R})$ | <b>No (spacetime emergent)</b>  | <b>No (probability emergent)</b> | <b>No (global condition)</b> | <b>Derived (Theorem 9.1)</b> |

Several points deserve elaboration. The Lagrangian and Hamiltonian formulations of mechanics emerge, as noted in Section 9, as effective descriptions in the smooth coherent limit of Phase Theory. The Lagrangian  $(\mathcal{L}_{\text{eff}})$  is not postulated but derived from the first variation of  $(\mathcal{I})$ , and the Hamiltonian  $(\mathcal{H}_{\text{eff}})$  arises as the Legendre transform of  $(\mathcal{L}_{\text{eff}})$  in the appropriate limit. The canonical commutation relations of quantum

mechanics emerge from the second variation of  $\mathcal{I}$ —the quadratic form over perturbations of a robustly admissible configuration (Section 18)—which defines the symplectic structure of the effective phase space.

The Hamiltonian constraint of general relativity—the statement that the total Hamiltonian of the gravitational field vanishes on-shell—corresponds in Phase Theory to the statement that admissible evolution within a topological sector satisfies  $\delta_\lambda \mathcal{I} = 0$ , which by Theorem 9.1 is precisely the stationarity condition. The specific form of the Hamiltonian constraint—as a combination of the ADM lapse and shift functions—emerges from the decomposition of the admissibility condition in the limit where the base space  $\mathcal{B}$  admits a foliation into spatial hypersurfaces, which is itself an emergent feature of configurations with appropriate coherence structure. The full derivation is given in Paper 8.

The relationship to the Wald entropy functional [11] merits brief discussion. The Wald entropy formula assigns an entropy to black hole horizons that generalizes the Bekenstein-Hawking area formula to higher-derivative gravity theories. The Wald entropy is defined as the Noether charge associated with diffeomorphism symmetry in the gravitational action. In Phase Theory, the stability term  $\mathcal{S}[\Phi]$  plays an analogous role: evaluated on configurations with emergent causal horizons,  $\mathcal{S}[\Phi]$  is the quantity that determines the admissibility of near-horizon configurations, and its variation with respect to horizon parameters reproduces the Wald entropy formula to leading order. This connection suggests that black hole entropy, in Phase Theory, is a measure of the stability margin of horizon configurations—a deeply geometric and physically intuitive result. The full derivation is deferred to Paper 8.

## 14. Comparison with Competing Quantum Gravity Frameworks

**(a) String Theory and the Landscape Problem.** String theory remains the most extensively developed approach to quantum gravity, producing an impressive degree of mathematical consistency and partial unification [12, 13]. However, the string landscape—the enormous collection of consistent vacuum configurations arising from flux compactification, estimated at approximately  $(10^{500})$  [14, 15]—poses a severe problem for the predictive content of the theory. In the absence of a selection mechanism that uniquely (or nearly uniquely) identifies the physical vacuum, string theory is in principle compatible with an enormous range of low-energy physics, and its predictive power for particle physics observables is correspondingly limited. Phase Theory's admissibility functional provides precisely the type of selection mechanism that the string landscape lacks: the axioms A1–A5, combined with the coherence bound and robust admissibility, reduce the space of physically realizable configurations to a discrete, finite set—approximately  $(25 \pm 10)$  stable species (Paper 3). Furthermore, the action principle used in string theory is postulated—the Nambu-Goto or Polyakov action is assumed from the outset. In Phase Theory, the action emerges from the first-order variation of  $(\mathcal{I})$  and is therefore derived, not assumed.

**(b) Loop Quantum Gravity.** Loop quantum gravity (LQG) achieves background independence and produces a discrete quantum geometry at the Planck scale through the construction of spin network states and spin foam amplitudes [16, 17, 18]. The discreteness of area and volume operators is a celebrated result of LQG. However, LQG achieves this discreteness by quantizing the gravitational field using loop variables—a process that discretizes spacetime geometry as an input assumption, in the sense that the Hilbert space is built over discrete spin network labels from the outset. Phase Theory, by contrast, obtains discreteness as a derived property of the

admissible configuration space (Theorem 11.1), without discretizing anything at the level of the base space  $\mathcal{B}$ . The base space need not be a lattice, a simplicial complex, or a spin foam: its topology is unconstrained, and discreteness emerges from the coherence constraint. Additionally, LQG retains an action principle—the Ashtekar-Barbero action [18]—as a fundamental input. In Phase Theory, this action emerges from the admissibility condition in the appropriate limit.

**(c) Causal Set Theory.** Causal set theory [19, 20, 21] proposes that the fundamental structure underlying spacetime is a discrete partial order—a causal set—from which the continuum causal structure of Lorentzian spacetime emerges in an appropriate limit. The approach has the virtues of background independence and a clear implementation of the causal principle. However, the causal set itself is postulated as the fundamental structure; its dynamics is typically specified by a path integral over causal sets with a postulated amplitude. In Phase Theory, causal ordering is an emergent relational structure among admissible phase configurations (Section 8, Paper 9): no discrete causal elements are postulated, and the causal partial order arises from the transitivity of admissible evolution (Theorem 8.1) combined with the time-asymmetry of the stability term. The dynamics requires no additional postulated amplitude: it follows from the admissibility condition alone.

**(d) Causal Dynamical Triangulations (CDT).** CDT [22, 23] implements a non-perturbative path integral over geometries using a discretization into simplicial manifolds, imposing a global causality constraint that restricts the geometries summed to those with a definite causal structure—a condition found to be necessary for the existence of a well-defined continuum limit. The causality constraint in CDT plays a structural role analogous to the admissibility condition in Phase Theory: in both cases, a global coherence-like requirement selects physically viable configurations. However, in CDT the

causality constraint and the path integral amplitude are both postulated externally; the simplicial discretization is a computational tool rather than a fundamental claim about the structure of space. In Phase Theory, the analog of the causality constraint is built intrinsically into the coherence term  $\mathcal{C}[\Phi]$ : configurations that would violate the emergent causal structure by having incoherent phase gradients in the time-like direction are automatically inadmissible, with no separate causality postulate required. The simplicial structure of CDT emerges in Phase Theory from the discrete admissible configuration space of Section 11, again without postulation.

## 15. Implications for Subsequent Papers

The results established in the present paper serve as the logical foundation for the derivations carried out in Papers 3 through 9 of the Phase Theory series. The following structured summary specifies precisely which results of Paper 2 underpin each subsequent paper.

| Paper   | Title                      | Foundational Dependencies from Paper 2                                                                                                                                                                                                                                                                                                                            |
|---------|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Paper 3 | Discrete Particle Spectrum | Theorem 11.1 (Spectral Discreteness) establishes that the spectrum of particle masses and charges is discrete. Theorem 7.1 (Topological Protection) and Corollary 7.1 establish that stable particles correspond to topologically charged admissible configurations. Definition 7.1 (Topological Charge) provides the classification scheme for particle species. |
| Paper 4 | Phase Photons and Light    | The coherence term analysis (§5.1, §6) identifies admissible propagating wave configurations—those satisfying $\ \nabla \mathcal{B}[\Phi]_{\bar{L}^2}\ $                                                                                                                                                                                                          |



| Paper   | Title                           | Foundational Dependencies from Paper 2                                                                                                                                                                                                                                                                                        |
|---------|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         |                                 | $\leq \kappa_c$ )—as the phase-theoretic photons. The coherence length (§6) determines the dispersion relation, and the holonomy structure (§6) determines polarization.                                                                                                                                                      |
| Paper 5 | Emergent Quantum Statistics     | Theorem 12.1 (Emergent Born Rule) is derived in full in Paper 5. The coarse-graining structure of §12 provides the framework within which the amplitude functional is defined and the Born rule limit is achieved.                                                                                                            |
| Paper 6 | Measurement Theory              | Theorem 10.1 (Ontological Elimination) is the key input: definite measurement outcomes arise because inadmissible branches are ontologically absent. The admissible branch structure (§10, Remark 2.2) replaces the collapse postulate. Section 12's discussion of the measurement problem provides the conceptual framework. |
| Paper 7 | Gauge Structure and Confinement | The topological sector decomposition (§7, §18) provides the classification of gauge sectors. Quantized topological charges (Definition 7.1) yield quantized gauge charges. The holonomy analysis of §6 generates the gauge connection. Corollary 7.1 establishes topological confinement.                                     |
| Paper 8 | Gravitational Dynamics          | The emergent action principle (§9, Theorem 9.1) is the foundation for the derivation of the Einstein-Hilbert action and its corrections. The stability term analysis (§5.3) is connected to the Wald                                                                                                                          |

| Paper   | Title                        | Foundational Dependencies from Paper 2                                                                                                                                                                                                                                                                                                      |
|---------|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         |                              | entropy functional (§13) and provides the admissibility condition near causal horizons.                                                                                                                                                                                                                                                     |
| Paper 9 | Cosmology and Temporal Order | Theorem 8.1 (Transitivity of Physical Evolution) establishes the partial order on admissible configurations from which temporal order emerges. The admissibility boundary analysis (§17) provides the framework for phase transitions and symmetry breaking. The stability term's time-asymmetry generates the thermodynamic arrow of time. |

16. Falsifiability Conditions

A theoretical framework is scientifically meaningful only to the extent that it generates predictions that, if experimentally disconfirmed, would falsify the theory. Phase Theory, despite its foundational character, is not exempt from this requirement. The following four falsifiability conditions identify the principal empirical stakes of the admissibility framework.

**Falsifiability Condition F1 (Structural Reproduction).**

If it can be demonstrated that no admissibility functional satisfying Axioms A1–A5 is capable of reproducing the observed spectrum of stable particles, coupling constants, and conservation laws of the Standard Model as derived quantities in Papers 3 and 7, then the admissibility framework as defined herein is falsified. This condition requires a mathematical no-go proof, not merely failure of a particular choice of  $\mathcal{M}_\phi$ .

### **Falsifiability Condition F2 (Discreteness).**

If physical observables—energy levels, charge values, angular momenta—are found to take values in a genuine continuum, unconstrained by any discreteness structure at any scale, then Theorem 11.1 is falsified and with it the admissibility framework. Any experimental detection of a fractional charge not predicted by the topological classification of Paper 3, or of an energy level that does not correspond to a robustly admissible configuration, constitutes evidence against the theory.

### **Falsifiability Condition F3 (No New Species Beyond Spectrum).**

Phase Theory predicts a finite number of stable particle species—approximately  $(25 \pm 10)$ , corresponding to the robustly admissible topological sectors of  $(\mathcal{M}_\phi)$ —as derived in Paper 3. If stable particles are discovered that cannot be accommodated within this topological classification, despite exhaustive attempts to identify the relevant topological sector, Phase Theory is falsified. This condition becomes testable as collider energies approach the spectral cutoff predicted by the coherence bound.

### **Falsifiability Condition F4 (Evolution Continuity).**

If physical evolution is observed to produce ontologically inadmissible intermediate configurations—if, for example, a physical process appears to require an intermediate state that Phase Theory classifies as nonexistent—then Definition 8.1 is falsified and with it the admissibility framework. In particular, if quantum erasure experiments reveal evolution paths through configurations that violate the coherence bound of Lemma 6.1, this would

constitute a direct falsification.

These conditions are not merely formal; they admit concrete experimental protocols. Falsifiability Condition F2 can be tested via precision spectroscopy at collider energies approaching the predicted spectral cutoff: as collision energies increase toward the energy scale corresponding to  $\ell_{\min}^{-1}$ , Phase Theory predicts a hard cutoff in the production of new stable species, rather than the continued proliferation expected in other frameworks. Condition F3 is tested simultaneously: the species count stabilizes at the topological limit, and any discovery of a new stable species above this limit would falsify the theory. Condition F4 admits tests via quantum erasure experiments with ever-increasing path length and decoherence time: if coherent interference is maintained over path lengths exceeding the coherence bound of Lemma 6.1, the bound is violated and the theory is challenged. The admissibility framework is therefore falsifiable in principle across multiple experimental fronts, satisfying the Popperian requirement for scientific theories [24].

## 17. The Admissibility Boundary and Phase Transitions

The admissibility boundary  $\partial \mathcal{A} = \mathcal{I}^{-1}(\{0\}) \subset \mathfrak{C}$  is the set of configurations that are physically admissible in the limiting sense: they satisfy  $\mathcal{I}[\Phi] = 0$ , neither clearly forbidden nor robustly admissible. These are the marginal configurations—configurations at the edge of physical existence. The analysis of  $\partial \mathcal{A}$  is important for understanding phase transitions, symmetry breaking, and the formation of topological defects.

**Theorem 17.1 (Measure of the Admissibility Boundary).**

The admissibility boundary  $\partial\mathcal{A}$  has measure zero in  $\mathfrak{C}$  under any natural functional measure compatible with Axiom A5 (Scale Neutrality). That is,  $\mu(\partial\mathcal{A}) = 0$  for any measure  $\mu$  on  $\mathfrak{C}$  that respects the scale invariance of  $\mathcal{I}$ .

### Proof Sketch.

By Axiom A5,  $\mathcal{I}$  is scale-neutral—it carries no dependence on an externally imposed scale. Any natural measure  $\mu$  on  $\mathfrak{C}$  compatible with A5 must therefore be absolutely continuous with respect to the  $(L^2)$  functional measure and invariant under scale transformations of  $\mathcal{B}$ . The level set  $\mathcal{I}^{-1}(0)$  is the zero level set of a functional that is (by the smoothness assumptions of Section 8) Fréchet differentiable at configurations where  $\delta\mathcal{I} \neq 0$ . By the implicit function theorem for Fréchet-differentiable functionals, the zero level set is a codimension-1 submanifold of  $\mathfrak{C}$  at regular points, and hence has measure zero under any absolutely continuous measure. At critical points where  $\delta\mathcal{I} = 0$ , the level set may have higher codimension or singular structure, but remains measure-zero by a standard capacity argument.

□

### Corollary 17.1.

Generic physical configurations are either robustly admissible ( $\mathcal{I} > 0$ ) or clearly forbidden ( $\mathcal{I} < 0$ ). Marginal configurations in  $\partial\mathcal{A}$  are non-generic: they form a measure-zero subset of  $\mathfrak{C}$  and are not encountered by generic physical evolution.

The role of  $\partial\mathcal{A}$  in Phase Theory's account of phase transitions is central. A *phase transition* in Phase Theory refers to a transition between distinct topological sectors—a process in which the topological charge  $Q(\Phi)$  changes. By Theorem 7.1, such transitions cannot occur via continuous admissible evolution. They require passage through the admissibility boundary: the evolving configuration must reach  $\partial\mathcal{A}$  (becoming marginal) before crossing into a configuration of different topological charge. The suppression of such transitions—the large free energy barrier associated with passing through  $\partial\mathcal{A}$ —corresponds to the stability of topological defects and the conservation of topological charge. First-order phase transitions in this language are those in which the transition rate through  $\partial\mathcal{A}$  is tunneling-suppressed (the marginal configurations are locally isolated); second-order phase transitions correspond to the case where  $\partial\mathcal{A}$  is continuously accessible and the topological charge changes smoothly through it, which requires a more refined treatment of the boundary structure beyond the leading-order analysis presented here.

Spontaneous symmetry breaking admits a natural description in the Phase Theory framework. A symmetry of  $\mathcal{I}[\Phi]$  corresponds to a continuous deformation of  $\mathcal{C}_{\mathcal{A}}$  that leaves  $\mathcal{I}$  invariant. If the maximally symmetric configuration lies on the admissibility boundary  $\partial\mathcal{A}$ , then the robustly admissible configurations of  $\mathcal{C}_{\mathcal{A}}$  do not share the full symmetry—they are asymmetric ground states. The admissibility boundary in the neighborhood of the symmetric configuration is a manifold on which  $\mathcal{I} = 0$ , and the robustly admissible configurations constitute a discrete set of minima at  $(\sigma > 0)$ . The vacuum manifold of the symmetry-broken phase corresponds to the orbit of a single robustly admissible configuration under the broken symmetry generators. Goldstone-like modes emerge as tangential deformations along  $\partial\mathcal{A}$

connecting distinct but marginally admissible configurations—modes that cost zero admissibility to first order, corresponding to the masslessness of Goldstone bosons.

The cosmological implications of  $\partial\mathcal{A}$  are explored in detail in Paper 9. During the earliest phases of cosmological evolution, when the coherence length was short and the phase configuration was highly energetic, the universe's configuration may have lain on or near  $\partial\mathcal{A}$ —the admissibility margin. Cosmological phase transitions—electroweak symmetry breaking, quark confinement—correspond in Phase Theory to the universe's configuration moving from a marginal or near-marginal state through  $\partial\mathcal{A}$  into a robustly admissible topological sector. The formation of topological defects (cosmic strings, monopoles) at such transitions corresponds to the inevitable creation of nontrivial topological charge when a configuration crosses  $\partial\mathcal{A}$ : configurations on different sides of the boundary in different spatial regions cannot be continuously connected, and the interfacial regions acquire nontrivial winding—precisely the homotopy-group argument for defect formation familiar from condensed matter physics [25].

## 18. Mathematical Structure of the Configuration Space

This section provides a rigorous characterization of the admissible configuration space  $\mathcal{C}_{\mathcal{A}} = \{\Phi \in \mathcal{C} : I[\Phi] \geq 0\}$  as a mathematical object, establishing results that underlie the geodesic and perturbative analyses used in all subsequent papers.

### Theorem 18.1 (Closedness of $\mathcal{C}_{\mathcal{A}}$ ).

$\mathcal{C}_{\mathcal{A}}$  is a closed subset of  $\mathcal{C}$  under the  $(L^2)$  norm topology. That is, if  $\{\Phi_n\}_{n \geq 1} \subset \mathcal{C}_{\mathcal{A}}$

$\mathfrak{C}_{\mathcal{A}}$  is a sequence with  $(\Phi_n \rightarrow \Phi)$  in  $(L^2)$ , then  $(\Phi \in \mathfrak{C}_{\mathcal{A}})$ .

### Proof Sketch.

Since each  $(\Phi_n \in \mathfrak{C}_{\mathcal{A}})$ , we have  $(\mathcal{I}[\Phi_n] \geq 0)$ . Under the stated regularity (Sobolev maps with square-integrable gradients), the topology terms  $(\mathcal{T}[\Phi_n])$  are continuous under  $(L^2)$  convergence (they are topological invariants that jump only at discontinuous limits), the stability terms  $(\mathcal{S}[\Phi_n])$  are lower-semicontinuous (stability can only be lost, not gained, under limits), and the coherence terms  $(\mathcal{C}[\Phi_n])$  are lower-semicontinuous by the lower-semicontinuity of the  $(L^2)$ -gradient norm under weak convergence. Therefore  $(\mathcal{I}[\Phi] \geq \liminf_{n \rightarrow \infty} \mathcal{I}[\Phi_n] \geq 0)$ , so  $(\Phi \in \mathfrak{C}_{\mathcal{A}})$ .

□

### Theorem 18.2 (Topological Sector Decomposition).

The admissible configuration space decomposes as a countable disjoint union of topological sectors:

$$\mathfrak{C}_{\mathcal{A}} = \bigsqcup_{Q \in \pi_n(\mathcal{M}_{\phi})} \mathfrak{C}_Q$$

where each sector  $\mathfrak{C}_Q = \{\Phi \in \mathfrak{C}_{\mathcal{A}} : Q(\Phi) = Q\}$  is path-connected within the admissible set—that is, any two configurations in  $\mathfrak{C}_Q$  can be connected by an admissible evolution path remaining within  $\mathfrak{C}_Q$ .



Each topological sector  $\mathcal{C}_Q$  carries a natural functional Riemannian metric induced by the admissibility functional. Define the inner product on the tangent space  $T_{\Phi}\mathcal{C}_Q$  (the space of admissible perturbations of  $\Phi$  within sector  $Q$ ) by

$$\langle \delta\Phi_1, \delta\Phi_2 \rangle_{\Phi} = -\delta^2\mathcal{I}[\Phi](\delta\Phi_1, \delta\Phi_2)$$

where  $\delta^2\mathcal{I}$  denotes the second Fréchet derivative of  $\mathcal{I}$  at  $\Phi$ , applied to the pair of perturbations  $(\delta\Phi_1, \delta\Phi_2)$ . The negative sign ensures positivity at robustly admissible configurations, where  $\Phi$  is a local maximum of  $\mathcal{I}$  within its sector (Section 9). This metric is well-defined at all robustly admissible configurations and degenerates at the admissibility boundary  $\partial\mathcal{A}$ , reflecting the loss of perturbative stability there. Physical evolution paths are, in the smooth coherent limit, geodesics on  $(\mathcal{C}_Q, \langle \cdot, \cdot \rangle)$ . The geodesic equation on this functional Riemannian manifold takes the form

$$\frac{D}{dt}\dot{\gamma}(t) = 0$$

where  $\frac{D}{dt}$  is the covariant derivative with respect to the metric  $\langle \cdot, \cdot \rangle$ , and  $\gamma(t)$  is the evolution path in  $\mathcal{C}_Q$ . In the smooth coherent limit, the Euler-Lagrange equations for the emergent action  $\mathcal{S}_{\text{eff}}$  of Section 9 emerge as the coordinate representation of this geodesic equation. Deviations from geodesic motion—non-linear corrections to the Euler-Lagrange equations—arise from higher-order terms in the expansion of  $\mathcal{I}$  around a robustly admissible configuration and correspond to loop corrections in quantum field theory.

The tangent space  $T_{\Phi}\mathcal{C}_Q$  at a robustly admissible configuration  $\Phi$  corresponds to the space of physical perturbations of  $\Phi$

$\Phi$ )—its fluctuations, excitations, and wave-like disturbances. In the limit where  $(\mathcal{B})$  has spacetime structure and  $(\Phi)$  is the vacuum configuration of a topological sector, this tangent space is infinite-dimensional and its elements are the modes of the quantum field associated with that sector. Diagonalizing the quadratic form  $(-\delta^2 \mathcal{I}[\Phi])$  on this space yields the normal mode decomposition familiar from free quantum field theory: the eigenvalues of the quadratic form are the squared masses of the excitation modes, and the eigenvectors are the mode functions. This correspondence between the second variation of  $(\mathcal{I})$  and the free-field mass spectrum is the mathematical bridge between Phase Theory and quantum field theory, and its implications for the Standard Model spectrum are developed in Paper 3.

## 19. Consistency with Known Physics

A foundational theory of physics must be consistent with all established experimental results in the regimes where those results have been validated. The admissibility framework of Phase Theory is constructed to reproduce known physics in appropriate limits. This section demonstrates the correspondence in three key cases.

**(a) Classical Mechanics.** Consider configurations  $(\Phi \in \mathcal{C}_Q)$  with large topological charge  $(|Q| \gg 1)$  and large coherence length  $(\ell_c(\Phi) \gg \ell_{\min})$ . In this limit, the relevant effective scale is the coherence length, which is macroscopic. The emergent action of Section 9 reduces, in this limit, to the classical action of a point particle or rigid body, and the geodesic equation on  $(\mathcal{C}_Q)$  reduces to Newton's second law or the Euler-Lagrange equations of classical mechanics. Specifically, the phase profile  $(\Phi)$  in this limit can be approximated by the configuration of a classical particle at position  $(q(t))$  with momentum  $(p = m \dot{q})$ , and the stationarity condition  $(\delta_\lambda \mathcal{I} = 0)$  reduces to  $(\ddot{q} = -\nabla V/m)$ . The classical

correspondence principle is not postulated but derived: it holds exactly in the limit of large topological charge and macroscopic coherence, which is precisely the regime where classical physics has been validated experimentally [26, 27].

**(b) Quantum Field Theory.** In the regime of small fluctuations around a robustly admissible configuration  $\phi_0$ , the admissibility functional can be expanded as

$$\mathcal{I}[\phi_0 + \delta\phi] = \mathcal{I}[\phi_0] + \frac{1}{2}\delta^2\mathcal{I}[\phi_0](\delta\phi, \delta\phi) + O(|\delta\phi|^3)$$

The leading correction is quadratic in the perturbation  $\delta\phi$ . Diagonalizing the quadratic form  $(-\delta^2\mathcal{I}[\phi_0])$  on the tangent space  $(T_{\phi_0}\mathcal{C}_Q)$  (Section 18) yields a free-field Hamiltonian of the form

$$\mathcal{H}_{\text{free}} = \sum_k \omega_k (a_k^\dagger a_k)$$

where  $(\omega_k)$  are the eigenvalues of the quadratic form and  $(a_k^\dagger, a_k)$  are creation and annihilation operators for the mode functions—the standard free-field Hamiltonian of quantum field theory [28, 29]. Interactions arise from the cubic and higher terms in the expansion of  $\mathcal{I}$ , whose coefficients determine the coupling constants of the effective field theory. The Standard Model interaction structure—the gauge symmetries, Yukawa couplings, and Higgs mechanism—emerges from the structure of  $\mathcal{I}$  at higher order, as derived in Paper 7. At this level of analysis, the connection to quantum field theory is exact in the perturbative regime: Phase Theory is consistent with all perturbative QFT predictions, with corrections at non-perturbative order controlled by the full non-linear admissibility structure.

**(c) General Relativity.** In the regime where the phase configuration  $\Phi$  exhibits emergent spacetime structure—a four-dimensional effective manifold  $\mathcal{M}^{(4)}$  with a Lorentzian metric  $(g_{\mu\nu})$  derived from the coherence structure of  $\Phi$ —the stability term  $\mathcal{S}[\Phi]$  evaluated on such configurations takes the form

$$\mathcal{S}[\Phi]_{\text{Big|}_{\text{spacetime}}} = \frac{1}{16\pi G} \int_{\mathcal{M}^{(4)}} R \sqrt{-g} \, d^4x + O(\ell_{\min}^2 R^2)$$

where  $R$  is the Ricci scalar of the emergent metric,  $G$  is Newton's constant (emergent from the phase manifold geometry), and the correction terms are suppressed by the square of the ratio of the minimum scale  $\ell_{\min}$  to the curvature radius. The stationarity condition  $\delta \lambda \mathcal{I} = 0$  then yields the Einstein field equations as the leading-order result, with Planck-suppressed corrections corresponding to higher-derivative gravity. The cosmological constant in this framework is not a free parameter but is determined by the zero-point structure of the admissible configuration space—its value is related to the density of robustly admissible configurations, and its computation is a principal goal of Paper 9. The consistency of Phase Theory with general relativity is thereby established at the level of the field equations, with all parameters (Newton's constant, the cosmological constant) emerging as derived quantities.

The present paper does not reproduce the full derivation of the Standard Model gauge group, the specific values of the three coupling constants, or the precise cosmological constant. These results require the detailed analysis of the phase manifold  $\mathcal{M}_\Phi$  carried out in Papers 7 and 8, where the specific homotopy groups and Riemannian geometry of  $\mathcal{M}_\Phi$  are fixed by the requirement that the emergent particle spectrum matches observation. What is established in the present paper is the framework within which these derivations are possible: the axiomatic structure, the canonical decomposition, and the formal correspondence with

known physics in appropriate limits. The framework is logically consistent with all established experimental results in the regimes where those results hold, and its novel predictions are confined to regimes—near the coherence scale  $\ell_{\min}$  and at the spectral cutoff—that are not yet experimentally accessible but are in principle testable.

## 20. Conclusion

This paper has established the formal structure of the phase admissibility functional  $\mathcal{I}[\Phi]$ , the central operative mechanism of Phase Theory. Beginning from the foundational ontology of Paper 1—in which phase  $\Phi: \mathcal{B} \rightarrow \mathcal{M}_{\phi}$  is the sole primitive of physical reality—the paper has motivated, defined, and analyzed the functional  $\mathcal{I}$  through five axioms (A1–A5), its canonical three-term decomposition  $\mathcal{I} = \mathcal{C} + \mathcal{T} + \mathcal{S}$ , and a sequence of formal theorems. The key results are: the coherence bound (Lemma 6.1), establishing an intrinsic minimum scale; topological protection (Theorem 7.1 and Corollary 7.1), establishing the absolute stability of topologically charged configurations; the transitivity of physical evolution (Theorem 8.1), providing the preorder from which time emerges; the emergent extremization principle (Theorem 9.1), deriving the action principle rather than postulating it; ontological elimination (Theorem 10.1), establishing that inadmissible configurations do not exist; spectral discreteness (Theorem 11.1), deriving quantization from admissibility; the emergent Born rule (Theorem 12.1, stated); and the structure theorems of the admissible configuration space (Theorems 18.1 and 18.2). Each of these results follows from the axioms without additional postulates, demonstrating the self-sufficiency of the admissibility framework.

The conceptual advances represented by this framework over traditional approaches are substantial. The action principle—the foundational object of all conventional physical theories from classical mechanics through general

relativity—is here a derived effective description, not a fundamental postulate. The Born rule—the probabilistic postulate of quantum mechanics that has resisted derivation for a century—emerges as a theorem in the appropriate limit. The discreteness of physical spectra—standardly postulated through canonical quantization—is derived from the geometry of the admissible configuration space. Inadmissible configurations are not merely suppressed or avoided but are ontologically absent—a distinction that resolves apparent paradoxes in quantum mechanics and quantum gravity at the foundational level. All of this is accomplished by a single mechanism—the admissibility functional—operating on the single primitive—phase—without exogenous postulates, background structures, or interpretation choices. The parsimony of the framework is not a cosmetic virtue but a logical necessity, given the foundational commitment of Phase Theory to phase as the unique ontological primitive.

Several important open questions remain. First, the precise determination of the phase target manifold  $\mathcal{M}_\phi$ —its dimension, topology, and Riemannian structure—is essential for translating the abstract results of this paper into concrete predictions. The specification of  $\mathcal{M}_\phi$  is the subject of Paper 3, where the requirement that the topological sectors yield the observed particle spectrum uniquely (up to a small class of equivalences) fixes the principal features of  $\mathcal{M}_\phi$ . Second, the explicit computation of  $\mathcal{I}[\Phi]$  in closed analytic form—beyond the decomposition into coherence, topological, and stability terms—remains an open problem of considerable difficulty. The decomposition provides a conceptual and qualitative framework, but the precise quantitative form of each term depends on the geometry of  $\mathcal{M}_\phi$  and the topology of  $\mathcal{B}$ , and may not admit a simple closed-form expression. Third, the connection between the admissibility functional and the renormalization group structure of effective field theories—the mechanism by which the effective action  $\mathcal{S}_{\text{eff}}$  runs

with energy scale—is not fully established in this paper and will require a detailed analysis of the higher-order terms in the variation of  $\mathcal{I}$ , to be carried out in Paper 5.

This paper occupies a unique position in the Phase Theory series: it is the operational bridge between the foundational ontology of Paper 1 and the concrete physical derivations of Papers 3 through 9. Every subsequent derivation—of the particle spectrum, of photon propagation, of gravitational dynamics, of quantum statistics, of the measurement process, of cosmological evolution—depends on the existence and properties of  $\mathcal{I}[\Phi]$  established here. The paper demonstrates that a radically parsimonious ontology—in which phase is the sole primitive—generates, through the single mechanism of global admissibility, a rich and highly constrained physical theory capable of reproducing the full phenomenological structure of modern physics in appropriate limits, while making novel, falsifiable predictions beyond those limits. The case presented here is mathematical in structure but foundational in aspiration: to show that the apparent complexity and plurality of physical law reduces, at the deepest level, to a single consistency criterion applied to a single entity. Whether this aspiration is fully realized depends on the theoretical and experimental work of the papers that follow.

---

## References

- [1] M. Hanks, "Phase Theory: A Unified Theory of Matter, Light, and Geometry," Preprint (2026). arXiv:ph.QG/2026.XXXXX [physics.gen-ph].
- [2] M. V. Berry, "Quantal phase factors accompanying adiabatic changes," Proc. R. Soc. London A **392**, 45–57 (1984).
- [3] S. Pancharatnam, "Generalized theory of interference and its applications," Proc. Indian Acad. Sci. A **44**, 247–262 (1956).
- [4] A. Shapere and F. Wilczek, eds., *Geometric Phases in Physics* (World Scientific, Singapore, 1989).

- [5] R. M. Wald, *General Relativity* (University of Chicago Press, Chicago, 1984).
- [6] R. Bousso and J. Polchinski, "Quantization of four-form fluxes and dynamical neutralization of the cosmological constant," J. High Energy Phys. **2000**, 006 (2000).
- [7] M. Schlosshauer, "Decoherence, the measurement problem, and interpretations of quantum mechanics," Rev. Mod. Phys. **76**, 1267 (2004).
- [8] D. Deutsch, "Quantum theory of probability and decisions," Proc. R. Soc. London A **455**, 3129–3137 (1999).
- [9] W. H. Zurek, "Environment-induced superselection rules," Phys. Rev. D **26**, 1862 (1982).
- [10] H. Everett III, "'Relative state' formulation of quantum mechanics," Rev. Mod. Phys. **29**, 454 (1957).
- [11] R. M. Wald, "Black hole entropy is the Noether charge," Phys. Rev. D **48**, R3427 (1993).
- [12] J. Polchinski, *String Theory, Volume I: An Introduction to the Bosonic String* (Cambridge University Press, Cambridge, 1998).
- [13] J. Polchinski, *String Theory, Volume II: Superstring Theory and Beyond* (Cambridge University Press, Cambridge, 1998).
- [14] S. Kachru, R. Kallosh, A. Linde, and S. P. Trivedi, "de Sitter vacua in string theory," Phys. Rev. D **68**, 046005 (2003).
- [15] L. Susskind, "The anthropic landscape of string theory," in *Universe or Multiverse?*, ed. B. Carr (Cambridge University Press, Cambridge, 2007), pp. 247–266.
- [16] C. Rovelli, *Quantum Gravity* (Cambridge University Press, Cambridge, 2004).
- [17] T. Thiemann, *Modern Canonical Quantum General Relativity* (Cambridge University Press, Cambridge, 2007).
- [18] A. Ashtekar, "New variables for classical and quantum gravity," Phys. Rev. Lett. **57**, 2244 (1986).
- [19] L. Bombelli, J. Lee, D. Meyer, and R. D. Sorkin, "Space-time as a causal set," Phys. Rev. Lett. **59**, 521 (1987).
- [20] R. D. Sorkin, "Causal sets: Discrete gravity," in *Lectures on Quantum Gravity*, ed. A. Gomberoff and D. Marolf (Springer, New York, 2005), pp. 305–327.



- [21] J. Henson, "The causal set approach to quantum gravity," in *Approaches to Quantum Gravity*, ed. D. Oriti (Cambridge University Press, Cambridge, 2009), pp. 393–413.
- [22] J. Ambjørn, J. Jurkiewicz, and R. Loll, "Dynamically triangulating Lorentzian quantum gravity," Nucl. Phys. B **610**, 347–382 (2001).
- [23] J. Ambjørn, J. Jurkiewicz, and R. Loll, "Emergence of a 4D world from causal quantum gravity," Phys. Rev. Lett. **93**, 131301 (2004).
- [24] K. R. Popper, *The Logic of Scientific Discovery* (Routledge, London, 1959).
- [25] T. W. B. Kibble, "Topology of cosmic domains and strings," J. Phys. A: Math. Gen. **9**, 1387 (1976).
- [26] C. W. Misner, K. S. Thorne, and J. A. Wheeler, *Gravitation* (W. H. Freeman, San Francisco, 1973).
- [27] A. Einstein, "Die Grundlage der allgemeinen Relativitätstheorie," Ann. Phys. **49**, 769–822 (1916).
- [28] S. Weinberg, *The Quantum Theory of Fields, Volume I: Foundations* (Cambridge University Press, Cambridge, 1995).
- [29] M. E. Peskin and D. V. Schroeder, *An Introduction to Quantum Field Theory* (Addison-Wesley, Reading, MA, 1995).
- [30] J. Zinn-Justin, *Quantum Field Theory and Critical Phenomena*, 4th ed. (Oxford University Press, Oxford, 2002).
- [31] K. G. Wilson and J. Kogut, "The renormalization group and the  $\epsilon$  expansion," Phys. Rep. **12**, 75–200 (1974).
- [32] L. P. Kadanoff, "Scaling laws for Ising models near  $(T_c)$ ," Physics **2**, 263–272 (1966).
- [33] M. F. Atiyah, "Topological quantum field theory," Publ. Math. IHÉS **68**, 175–186 (1988).
- [34] E. Witten, "Topological quantum field theory," Commun. Math. Phys. **117**, 353–386 (1988).
- [35] A. Einstein, B. Podolsky, and N. Rosen, "Can quantum-mechanical description of physical reality be considered complete?" Phys. Rev. **47**, 777 (1935).
-

